

HW 5

CSE 331

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Due: **Saturday** November 2nd, 2024 11:00pm

Task 1: Twice to Meet You

(a) prove by calculation that $\text{sum}(L) = a + b$.

Solution:

$\text{sum}(L) = \text{sum}(a :: b :: \text{nil})$	since $L = a :: b :: \text{nil}$
$= a + \text{sum}(b :: \text{nil})$	def of sum
$= a + b + \text{sum}(\text{nil})$	def of sum
$= a + b$	def of sum

(b) prove by calculation that $\text{sum}(\text{twice-evens}(L)) = 2a + b$.

Solution:

$\text{sum}(\text{twice-evens}(L)) = \text{sum}(\text{twice-evens}(a :: b :: \text{nil}))$	since $L = a :: b :: \text{nil}$
$= \text{sum}(2a :: \text{twice-odds}(b :: \text{nil}))$	def of twice-evens
$= \text{sum}(2a :: b :: \text{twice-evens}(\text{nil}))$	def of twice-odds
$= \text{sum}(2a :: b :: \text{nil})$	def of twice-evens
$= 2a + \text{sum}(b :: \text{nil})$	def of sum
$= 2a + b + \text{sum}(\text{nil})$	def of sum
$= 2a + b$	def of sum

(c) prove by calculation that $\text{sum}(\text{twice-odds}(L)) = a + 2b$.

Solution:

$\text{sum}(\text{twice-odds}(L)) = \text{sum}(\text{twice-odds}(a :: b :: \text{nil}))$	since $L = a :: b :: \text{nil}$
$= \text{sum}(a :: \text{twice-evens}(b :: \text{nil}))$	def of twice-odds
$= \text{sum}(a :: 2b :: \text{twice-odds}(\text{nil}))$	def of twice-evens
$= \text{sum}(a :: 2b :: \text{nil})$	def of twice-odds
$= a + \text{sum}(2b :: \text{nil})$	def of sum
$= a + 2b + \text{sum}(\text{nil})$	def of sum
$= a + 2b$	def of sum

- (d) Prove that the code is correct by showing that $\text{sum}(\text{twice-evens}(L)) + \text{sum}(\text{twice-odds}(L)) = 3 \text{sum}(L)$

Solution:

$$\begin{aligned}
 \text{sum}(\text{twice-evens}(L)) + \text{sum}(\text{twice-odds}(L)) &= 2a + b + \text{sum}(\text{twice-odds}(L)) && \text{part (b)} \\
 &= 2a + b + a + 2b && \text{part (c)} \\
 &= 3(a + b) && \text{algebra} \\
 &= 3 \text{sum}(L) && \text{part (a)}
 \end{aligned}$$

Task 2: Swap 'Til You Drop

- (a) Let x be an integer. Prove that $\text{swap}(\text{swap}(x :: \text{nil})) = x :: \text{nil}$

Solution:

$$\begin{aligned}
 \text{swap}(\text{swap}(x :: \text{nil})) &= \text{swap}(x :: \text{nil}) && \text{def of swap} \\
 &= x :: \text{nil} && \text{def of swap}
 \end{aligned}$$

- (b) Let x and y be integers and R be a list. Prove that $\text{swap}(\text{swap}(x :: y :: R)) = x :: y :: \text{swap}(\text{swap}(R))$

Solution:

$$\begin{aligned}
 \text{swap}(\text{swap}(x :: y :: R)) &= \text{swap}(y :: x :: \text{swap}(R)) && \text{def of swap} \\
 &= x :: y :: \text{swap}(\text{swap}(R)) && \text{def of swap}
 \end{aligned}$$

- (c) Let a, b, c, d, e be integers and list $L = a :: b :: c :: d :: e :: \text{nil}$. Prove that $\text{swap}(\text{swap}(L)) = L$.

Solution:

$$\begin{aligned}
 \text{swap}(\text{swap}(L)) &= \text{swap}(\text{swap}(a :: b :: c :: d :: e :: \text{nil})) && L = a :: b :: c :: d :: e :: \text{nil} \\
 &= a :: b :: \text{swap}(\text{swap}(c :: d :: e :: \text{nil})) && \text{part (b) with } x = a, y = b \\
 &= a :: b :: c :: d :: \text{swap}(\text{swap}(e :: \text{nil})) && \text{part (b) with } x = c, y = d \\
 &= a :: b :: c :: d :: e :: \text{nil} && \text{part (a) with } x = e
 \end{aligned}$$

- (d) Prove that the code is correct by showing that $\text{swap}(\text{swap}(\text{cons}(1, \text{cons}(2, L)))) = \text{cons}(1, \text{cons}(2, L))$

Solution: Using notations: $\text{cons}(1, \text{cons}(2, L)) = 1 :: 2 :: L$, where $L = c :: d :: e :: \text{nil}$ for some integers c, d , and e , we have:

$$\begin{aligned}
 \text{swap}(\text{swap}(\text{cons}(1, \text{cons}(2, L)))) &= \text{swap}(\text{swap}(1 :: 2 :: L)) \\
 &= \text{swap}(\text{swap}(1 :: 2 :: c :: d :: e :: \text{nil})) && L = c :: d :: e :: \text{nil} \\
 &= 1 :: 2 :: c :: d :: e :: \text{nil} && \text{part (c) with } a = 1, b = 2 \\
 &= 1 :: 2 :: L && L = c :: d :: e :: \text{nil} \\
 &= \text{cons}(1, \text{cons}(2, L))
 \end{aligned}$$

Task 3: I Neg To Differ

(a) repeat definition of ns from HW4

Solution:

$$ns : \mathbb{Z} \rightarrow \mathbb{Z}$$

$$\begin{aligned} ns(j) &:= 18 - j && \text{if } 0 \leq j \leq 5 \text{ or } 13 \leq j \leq 18 \\ ns(j) &:= 31 - j && \text{if } 6 \leq j \leq 12 \text{ or } 19 \leq j \leq 25 \\ ns(j) &:= j && \text{if } j < 0 \text{ or } j > 25 \end{aligned}$$

(b) Let j be an integer. Prove by cases that $ns(ns(j)) = j$.

Solution: Let j be any integer. We argue by case on j :

- Case 1: suppose that $0 \leq j \leq 5$ or $13 \leq j \leq 18$

$$\begin{aligned} ns(ns(j)) &= ns(18 - j) && \text{def of ns} \\ &= 18 - (18 - j) && \text{def of ns, } 13 \leq 18 - j \leq 18 \text{ or } 0 \leq 18 - j \leq 5 \\ &= j \end{aligned}$$

- Case 2: suppose that $6 \leq j \leq 12$ or $19 \leq j \leq 25$

$$\begin{aligned} ns(ns(j)) &= ns(31 - j) && \text{def of ns} \\ &= 31 - (31 - j) && \text{def of ns, } 19 \leq 31 - j \leq 25 \text{ or } 6 \leq 31 - j \leq 12 \\ &= j \end{aligned}$$

- Case 3: suppose that $j < 0$ or $j > 25$

$$\begin{aligned} ns(ns(j)) &= ns(j) && \text{def of ns} \\ &= j && \text{def of ns} \end{aligned}$$

We have proven the claim in all cases. Since the cases are exhaustive, we have proven it in general.

Task 4: Fish and Skips

- a. Prove, by structural induction, that $\text{skip}(\text{echo}(S)) = S$ for any list S .

Solution: Define $P(S)$ to be the claim that $\text{skip}(\text{echo}(S)) = S$ for any list S . We will prove the claim by structural induction.

- Base Case (nil). We can calculate

$$\begin{aligned}\text{skip}(\text{echo}(\text{nil})) &= \text{skip}(\text{nil}) && \text{def of echo} \\ &= \text{nil} && \text{def of skip}\end{aligned}$$

- Inductive Hypothesis. Suppose that $P(L)$ holds for a list L . I.e., suppose that $\text{skip}(\text{echo}(L)) = L$.
- Inductive Step. We need to show $P(a :: L)$ for any element a that fits into L .

$$\begin{aligned}\text{skip}(\text{echo}(a :: L)) &= \text{skip}(a :: a :: \text{echo}(L)) && \text{def of echo} \\ &= \text{skip}(a :: a :: L) && \text{Inductive Hypothesis} \\ &= \text{keep}(a :: L) && \text{def of skip} \\ &= a :: L && \text{def of keep}\end{aligned}$$

- Conclusion. $P(S)$ holds for any list S by structural induction.

- b. Prove that $\text{skip}(1 :: 2 :: \text{echo}(L)) = 2 :: L$.

Solution:

$$\begin{aligned}\text{skip}(1 :: 2 :: \text{echo}(L)) &= \text{keep}(2 :: \text{echo}(L)) && \text{def of skip} \\ &= 2 :: \text{skip}(\text{echo}(L)) && \text{def of keep} \\ &= 2 :: L && \text{part (a)}\end{aligned}$$

Task 5: Sum As You Are

- (a) Define $f : \text{List}(\mathbb{Z}) \rightarrow \mathbb{Z}$ so that $f(xs)$ calculates the expression $\text{sum}(\text{mult}(\text{zip}(xs, \text{tail}(xs)))) + \text{sum}(xs)$.

Solution: Let L be $\text{List}(\mathbb{Z})$. Then we define:

$$\begin{aligned}f : \text{List}(\mathbb{Z}) &\rightarrow \mathbb{Z} \\ f(\text{nil}) &:= 0 \\ f(a :: \text{nil}) &:= a \\ f(a :: b :: L) &:= a \times b + f(b :: L) + a\end{aligned}$$

- (b) Prove by structural induction that $f(xs) = \text{sum}(\text{mult}(\text{zip}(xs, \text{tail}(xs)))) + \text{sum}(xs)$

Solution:

Define $P(S)$ to be the claim that $f(S) = \text{sum}(\text{mult}(\text{zip}(S, \text{tail}(S)))) + \text{sum}(S)$ for any $S : \text{List}(\mathbb{Z})$. We will prove the claim by structural induction.

- Base Cases.

Base Case 1: $S = \text{nil}$. We can calculate

$$\begin{aligned}
& \text{sum}(\text{mult}(\text{zip}(\text{nil}, \text{tail}(\text{nil})))) + \text{sum}(\text{nil}) \\
= & \text{sum}(\text{mult}(\text{zip}(\text{nil}, \text{nil}))) + \text{sum}(\text{nil}) && \text{def of tail} \\
= & \text{sum}(\text{mult}(\text{nil})) + \text{sum}(\text{nil}) && \text{def of zip} \\
= & \text{sum}(\text{nil}) + \text{sum}(\text{nil}) && \text{def of mult} \\
= & 0 + \text{sum}(\text{nil}) && \text{def of sum} \\
= & 0 + 0 && \text{def of sum} \\
= & 0 \\
= & f(\text{nil}) && \text{def of } f
\end{aligned}$$

Base Case 2: $S = a :: \text{nil}$ for some $a \in \mathbb{Z}$, i.e. the length of S is 1. We can calculate

$$\begin{aligned}
& \text{sum}(\text{mult}(\text{zip}(S, \text{tail}(S)))) + \text{sum}(S) \\
= & \text{sum}(\text{mult}(\text{zip}(a :: \text{nil}, \text{tail}(a :: \text{nil})))) + \text{sum}(a :: \text{nil}) && \text{Since } S = a :: \text{nil} \\
= & \text{sum}(\text{mult}(\text{zip}(a :: \text{nil}, \text{nil}))) + \text{sum}(a :: \text{nil}) && \text{def of tail} \\
= & \text{sum}(\text{mult}(\text{nil})) + \text{sum}(a :: \text{nil}) && \text{def of zip} \\
= & \text{sum}(\text{nil}) + \text{sum}(a :: \text{nil}) && \text{def of mult} \\
= & 0 + \text{sum}(a :: \text{nil}) && \text{def of sum} \\
= & 0 + a && \text{def of sum} \\
= & a \\
= & f(a :: \text{nil}) && \text{def of } f
\end{aligned}$$

- Inductive Hypothesis. Suppose that $P(L)$ and $P(b :: L)$ holds for a list L and a $b \in \mathbb{Z}$. I.e., suppose that $f(L) = \text{sum}(\text{mult}(\text{zip}(L, \text{tail}(L)))) + \text{sum}(L)$, and $f(b :: L) = \text{sum}(\text{mult}(\text{zip}(b :: L, \text{tail}(b :: L)))) + \text{sum}(b :: L)$.
- Inductive Step. We need to show $P(a :: b :: L)$ for any $a, b \in \mathbb{Z}$.

$$\begin{aligned}
& \text{sum}(\text{mult}(\text{zip}(a :: b :: L, \text{tail}(a :: b :: L)))) + \text{sum}(a :: b :: L) \\
= & \text{sum}(\text{mult}(\text{zip}(a :: b :: L, b :: L))) + \text{sum}(a :: b :: L) && \text{def of tail} \\
= & \text{sum}(\text{mult}((a, b) :: \text{zip}(b :: L, L))) + \text{sum}(a :: b :: L) && \text{def of zip} \\
= & \text{sum}(\text{mult}((a, b) :: \text{zip}(b :: L, \text{tail}(b :: L))) + \text{sum}(a :: b :: L) && \text{def of tail} \\
= & \text{sum}(a \times b :: \text{mult}(\text{zip}(b :: L, \text{tail}(b :: L)))) + \text{sum}(a :: b :: L) && \text{def of mult} \\
= & a \times b + \text{sum}(\text{mult}(\text{zip}(b :: L, \text{tail}(b :: L)))) + \text{sum}(a :: b :: L) && \text{def of sum} \\
= & a \times b + \text{sum}(\text{mult}(\text{zip}(b :: L, \text{tail}(b :: L)))) + a + \text{sum}(b :: L) && \text{def of sum} \\
= & a \times b + f(b :: L) + a && \text{Inductive Hypothesis} \\
= & f(a :: b :: L) && \text{def of } f
\end{aligned}$$

- Conclusion. $P(S)$ holds for any list S by structural induction.